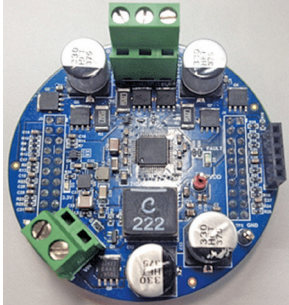
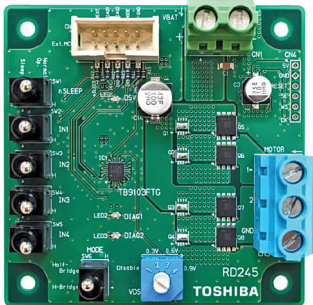


Interesting Reference Designs Of AUTOMOTIVE MOTOR DRIVE

Title of Reference Design	Automotive BLDC Motor Drive	Automotive Brushed DC Motor Control
Image		
Key Attractions	The reference design helps create 12V motor control systems with or without sensors. It handles motors up to 200W and offers features like noise reduction, long lifespan, and flexibility for different uses.	The reference design makes motor control easy and reliable, offering flexible options for driving motors safely. With built-in safety features, voltage protection, and stable power, it simplifies testing, making it suitable for automotive systems.
Highlights	The TIDA-00901 reference design by Texas Instruments (TI) enables design engineers to develop 3-phase brushless DC motor control systems for automotive applications. It supports sensorless commutation and Hall sensor integration, allowing adaptation to various motor configurations. Capable of driving motors up to 200W and handling currents up to 20A, it is suitable for 12V systems. The design includes the DRV8305-Q1 gate driver for motor control, the LM56300-Q1 step-down converter for power management, the LMT86-Q1 temperature sensor for thermal monitoring, and the DRV5013-Q1 Hall effect sensor for motor position feedback. Engineers can use its wide supply range, reverse battery protection, and control interface to address application requirements. The design provides ease of use, reliability, and adaptability for automotive applications, enabling efficient operation, long service life, and reduced noise.	Toshiba's reference design enables engineers to explore two motor control configurations: driving two motors in half-bridge mode or a single motor in H-bridge mode. Key features like dead time control, integrated fault detection, and fault indication outputs enhance motor reliability, safety, and durability. At the same time, the design's wide input voltage range makes it adaptable to various automotive systems. The design simplifies evaluating and testing the TB9103FTG's capabilities, allowing motor control via manual switches or an external microcontroller (MCU). A reverse connection protection circuit using the XPH1R104PS MOSFET safeguards against incorrect power supply polarity. An integrated voltage regulator IC ensures a stable 5V power supply for the design and external MCU. At the same time, the TB9005FNG enhances reliability with low-voltage reset, power-on reset, and a watchdog timer for optimal MCU performance.
Applications	The reference design suits automotive applications, such as HVAC blowers, water pumps, eTurbo systems, radiator fans, and oil pumps.	The reference design is suitable for latching and driving motors like doors, seats, and windows in automotive applications.
OEM Brand	Texas Instruments (TI)	Toshiba
URL	https://www.electronicsforu.com/electronics-projects/automotive-bldc-motor-drive-reference-design	https://www.electronicsforu.com/electronics-projects/automotive-brushed-dc-motor-control-reference-design